Hands-On-Activity No. 4.4

Hands-on Activity 4.4: Characters and Strings

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed: August 25, 2026**
Section: CPE11S2	**Date Submitted: August 25, 2026**
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1. Objective(s)

This activity aims to develop programming skills using characters and strings.

2. Intended Learning Outcomes

2.1 Use functions of the character handling library (ctype)

2.2 Use string and character input/output functions of the standard input/output library (stdio)

2.3 Use the string conversion functions of the general utilities library (stdlib)

2.4 Use string processing functions of the string handling library (string)

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1.
```
#include <iostream>
#include <cctype> //needed for islower and toupper
using namespace std;

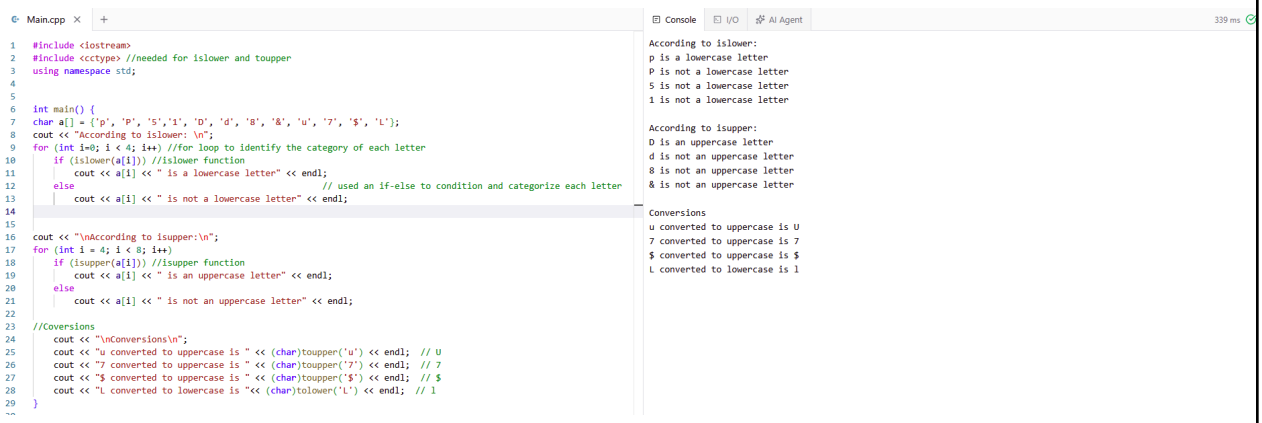
int main() {
char a[] = {'p', 'P', '5','1', 'D', 'd', '8', '&', 'u', '7', '$', 'L'};
cout << "According to islower: \n";
for (int i=0; i < 4; i++) //for loop to identify the category of each letter
   if (islower(a[i])) //islower function
      cout << a[i] << " is a lowercase letter" << endl;
   else                                  // used an if-else to condition and categorize each letter
      cout << a[i] << " is not a lowercase letter" << endl;

cout << "\nAccording to isupper:\n";
for (int i = 4; i < 8; i++)
   if (isupper(a[i])) //isupper function
      cout << a[i] << " is an uppercase letter" << endl;
   else
      cout << a[i] << " is not an uppercase letter" << endl;

//Coversions
   cout << "\nConversions\n";
```

```
        cout << "u converted to uppercase is " << (char)toupper('u') << endl;  // U
        cout << "7 converted to uppercase is " << (char)toupper('7') << endl;  // 7
        cout << "$ converted to uppercase is " << (char)toupper('$') << endl;  // $
        cout << "L converted to lowercase is "<< (char)tolower('L') << endl;  // l
    }
```

Screenshot:



2.
```
#include <stdio.h>
#include <ctype.h>
#include <cctype>
int main(void) {
    char c;

    printf("Enter a character: ");
    scanf(" %c", &c);

    printf("\n--- Character Analysis ---\n", c);

    // Inspection functions
    printf("isalpha:   %s\n", isalpha(c)   ? "True (letter)" : "False");
    printf("isdigit:   %s\n", isdigit(c)   ? "True (digit)" : "False");
    printf("isalnum:   %s\n", isalnum(c)   ? "True (alphanumeric)" : "False");
    printf("islower:   %s\n", islower(c)   ? "True (lowercase letter)" : "False");
    printf("isupper:   %s\n", isupper(c)   ? "True (uppercase letter)" : "False");
    printf("isspace:   %s\n", isspace(c)   ? "True (whitespace)" : "False");
    printf("ispunct:   %s\n", ispunct(c)   ? "True (punctuation)" : "False");
    printf("isxdigit:  %s\n", isxdigit(c)  ? "True (hexadecimal digit)" : "False");
    printf("print:     %s\n", isprint(c)   ? "True (printable character)" : "False");
```

```
    return 0;
}
```

Screenshot:

```c
#include <stdio.h>
#include <ctype.h>
#include <cctype>
int main(void) {
    char c;

    printf("Enter a character: ");
    scanf(" %c", &c);

    printf("\n--- Character Analysis ---\n", c);

    // Inspection functions
    printf("isalpha:   %s\n", isalpha(c)   ? "True (letter)" : "False");
    printf("isdigit:   %s\n", isdigit(c)   ? "True (digit)" : "False");
    printf("isalnum:   %s\n", isalnum(c)   ? "True (alphanumeric)" : "False");
    printf("islower:   %s\n", islower(c)   ? "True (lowercase letter)" : "False");
    printf("isupper:   %s\n", isupper(c)   ? "True (uppercase letter)" : "False");
    printf("isspace:   %s\n", isspace(c)   ? "True (whitespace)" : "False");
    printf("ispunct:   %s\n", ispunct(c)   ? "True (punctuation)" : "False");
    printf("isxdigit:  %s\n", isxdigit(c)  ? "True (hexadecimal digit)" : "False");
    printf("print:     %s\n", isprint(c)   ? "True (printable character)" : "False");

    return 0;
}
```

```
Enter a character:
i

--- Character Analysis ---
isalpha:   True (letter)
isdigit:   False
isalnum:   True (alphanumeric)
islower:   True (lowercase letter)
isupper:   False
isspace:   False
ispunct:   False
isxdigit:  False
print:     True (printable character)
```

3.
```c
#include <stdio.h>
#include <stdlib.h>

int main(void) {
    char str1[100], str2[100], str3[100], str4[100];
    int num1, num2, num3, num4;
    int total;

    // Prompt user and input 4 strings
    printf("Enter 4 integers as strings: :\n");
    scanf("%99s %99s %99s %99s", str1, str2, str3, str4);

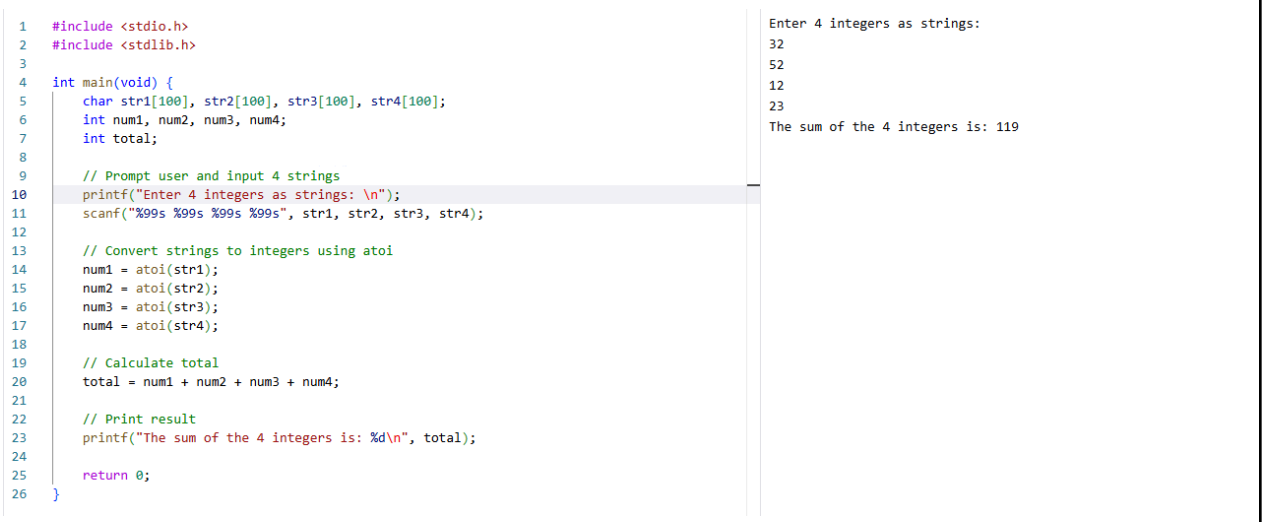
    // Convert strings to integers using atoi
    num1 = atoi(str1);
    num2 = atoi(str2);
    num3 = atoi(str3);
    num4 = atoi(str4);

    // Calculate total
    total = num1 + num2 + num3 + num4;

    // Print result
    printf("The sum of the 4 integers is: %d\n", total);
```

```
        return 0;
    }
```

Screenshot:



```c
#include <stdio.h>
#include <stdlib.h>

int main(void) {
    char str1[100], str2[100], str3[100], str4[100];
    int num1, num2, num3, num4;
    int total;

    // Prompt user and input 4 strings
    printf("Enter 4 integers as strings: \n");
    scanf("%99s %99s %99s %99s", str1, str2, str3, str4);

    // Convert strings to integers using atoi
    num1 = atoi(str1);
    num2 = atoi(str2);
    num3 = atoi(str3);
    num4 = atoi(str4);

    // Calculate total
    total = num1 + num2 + num3 + num4;

    // Print result
    printf("The sum of the 4 integers is: %d\n", total);

    return 0;
}
```

```
Enter 4 integers as strings:
32
52
12
23
The sum of the 4 integers is: 119
```

5. Conclusion

Learning more about features of cpp, characters, and strings.

6. Rubric

SO7